

INTERNSHIP COMPLETION

REPORT

Report Prepared By: - Manohar kothali

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Domain: - Embedded System Development

Designation in Team: - Intern

Report Submission Date: - 25/12/2025

Team Leader Name: - Sanvi kotha

Team Leader Contact No: - +91 9618740675

Report Send To: - Avi Kulkarni Sir (CEO) & Aditya Sir (HR)

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INTERN DECLARATION LETTER FOR INTERNSHIP

COMPLETION CERTIFICATE

To,

The Founder & CEO, & The HR Manager,

Blue Planet Infosolutions Pvt Ltd

55, ANMOL BLDG., PRASHANT SOC., PAUD ROAD BEHIND KRISHNA HOSPITAL, PUNE-38, HOSPITAL, PUNE-38, Maharashtra, India.

Date: 25/12/2025

Subject: Declaration and Request for Issuance of Internship Completion Certificate After Completion of Internship

Respected Sir/Madam,

I, **Manohar kothali**, currently working as an Intern in the **Embedded System Development** Team at **Blue Planet Infosolutions Pvt Ltd**, would like to express my sincere gratitude for the opportunity to be a part of this esteemed organization. My internship commenced on 5 June 2025 and is scheduled to conclude on 10 December 2025.

During my internship, I have gained valuable practical experience, technical knowledge, and professional exposure under the guidance and support of my mentors and team members. Working in this organization has greatly enhanced my skills, discipline, and understanding of workplace ethics and teamwork.

As my internship period is nearing completion, I hereby declare that before leaving the internship, I will complete all the ongoing tasks assigned to me and sincerely fulfill all my responsibilities up to the last working day of my internship. I will also ensure a proper handover of my work and responsibilities to my designated colleague or team member to maintain smooth continuity of the department's operations.

Additionally, I will submit a comprehensive report of all the tasks and projects I have completed during my six-month internship, along with all relevant proofs, documents, and records of work as required by the company. For verification purposes, I am also attaching a copy of my internship offer letter below this application as proof of the commencement of my internship at **Blue Planet Infosolutions Pvt Ltd**.

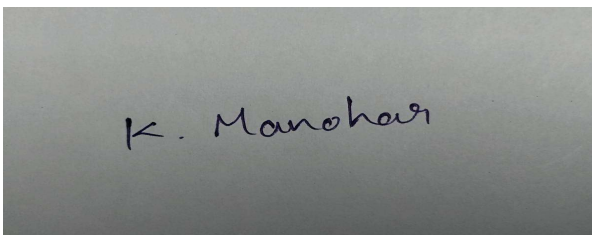
I hereby declare that all the documents, proofs, and information provided by me are true and valid to the best of my knowledge. In case any data, record, or document is later found to be invalid or misleading, I fully understand and agree that the company reserves the right to withhold or cancel the issuance of my Internship Completion Certificate without any objection from my side.

I therefore humbly request you to kindly issue my Internship Completion Certificate upon the successful completion of my internship tenure. This certificate will serve as an official record of my internship experience and learning at your organization.

I shall remain thankful for your kind support, guidance, and consideration throughout this period.

My Current Task Handover To :- Anishka

With sincere regards,

A rectangular box containing a handwritten signature in black ink that reads "K. Manohar".

Your Full Name: - Manohar kothali

Intern – Embedded System Development

Contact Number: - +91 9391457679

Email ID: - kothalimanohar01@gmail.com

INTERNSHIP TASK REPORT

Sr.No.	Task Assign Name	Task Assign Date	Tasks Completion Date	Task Assign By
1)	Design, Simulation, Sensor Research, PCB Development and Prototype Planning of Smart QR-based Pillbox System	6 th Jun 2025	29 th Jun 2025	Avi Kulkarni Sir
2)	Development and Prototyping of American Version Smart QR Pill Box with Advanced Features	1 th Jul 2025	30 th Jul 2025	Avi Kulkarni Sir
3)	PCB Design, Hardware Verification, and Research on LLM/SLM Integration for American Version QR Pill Box	1 st Aug 2025	19 th Aug 2025	Avi Kulkarni Sir
4)	PCB Hardware Checking, Manual PCB Fabrication, and Research on PCB Design Methods for American Version Pill Box	21 st Aug 2025	29 th Aug 2025	Avi Kulkarni Sir
5)	PCB Assembly, ESP32 Integration, Hardware Testing, and Real-Time Validation for American Version Smart Pill Box	2 nd Sep 2025	18 th Sep 2025	Avi Kulkarni Sir
6)	PCB Hardware Assembly, Demonstration, Extension Planning, and Real-Time Testing for Smart Pill Box	20 rd Sep 2025	30 th Sep 2025	Avi Kulkarni Sir

Sr.No.	Task Assign Name	Task Assign Date	Tasks Completion Date	Task Assign By
7)	3D Design Development, PCB Hardware Assembly Preparation, and Component Research for American Version Smart Pill Box	1 st Oct 2025	18 th Oct 2025	Avi Kulkarni Sir
8)	Battery Integration and 3D Design Development for American Version Smart Pill Box	19 th Oct 2025	28 th Oct 2025	Avi Kulkarni Sir
9)	Final Assembly and New PCB Design for American Version Smart Pill Box	1 st Nov 2025	8 st Nov 2025	Avi Kulkarni Sir
10)	New PCB Design (10-Slot), Product Testing, Explanation, and Courier Coordination	1 st Dec 2025	25 st Dec 2025	Avi Kulkarni Sir

WORK REPORT

- **Task Assign No:** - 1
 - **Task Assign Month:** - June
 - **Team Name:** - Embedded System Development
 - **Task Name:** - Design, Simulation, Sensor Research, PCB Development and Prototype Planning of Smart QR-based Pillbox System
 - **Task Assign By:** - Avi Kulkarni Sir
-
- **Work Description:** - Worked on end-to-end development of a Smart QR-based Pillbox system as part of the Embedded System Development team at Pan Health. The work included designing and simulating multiple pillbox configurations (6-way, 10-way, button-matrix based, touch-sensor based) using Arduino Uno, Arduino Nano, and ESP32 microcontrollers. Implemented Bluetooth and WiFi communication for real-time pill status monitoring. Conducted in-depth sensor research to identify suitable sensors for American-style pillbox bottle detection and prepared structured technical documentation. Designed and simulated multi-I2C sensor systems, data logging, alert mechanisms, and supporting embedded projects such as a Smart Plant Monitoring System.

Worked on PCB design for open and closed circuit configurations of the smart pillbox and planned the hardware prototype, including component selection and online procurement.

Additionally, reviewed teammates' reports, corrected circuits and code, suggested useful features, and reported progress to the project mentor.
-
- **Your Role in this Task:** - Embedded system designer, circuit simulator, sensor researcher, PCB designer, documentation contributor, and technical reviewer. Actively involved in both **individual development and team collaboration**.

- **What new you do in this Task:** - I integrated multiple technologies into a single healthcare-oriented embedded solution by:
 - Designing scalable pillbox architectures (6-way to 10-way)
 - Implementing button matrix and touch-based pill detection
 - Using ESP32 for wireless communication
 - Researching and comparing sensors for real medical use cases
 - Designing PCB layouts suitable for compact products
 - Reviewing and improving team documentation and designs

- **What new thing you learn from this Task:** - Learned the **complete embedded product development lifecycle**, including:
 - Circuit design and simulation using Proteus
 - Arduino and ESP32 programming
 - Bluetooth and WiFi communication
 - I2C protocol and multi-sensor interfacing
 - Sensor selection for medical devices
 - PCB design fundamentals and prototype planning
 - Technical documentation, reporting, and teamwork in a professional environment

- **Task Assign No:** - 2
 - **Task Assign Month:** - July
 - **Team Name:** - Embedded System Development
 - **Task Name:** - Development and Prototyping of American Version Smart QR Pill Box with Advanced Features
 - **Task Assign By:** - Avi Kulkarni Sir
-
- **Work Description:** - Worked continuously on the **design, development, and prototyping of the American version of the Smart QR Pill Box** as part of the Pan Health Embedded Development team. The work involved planning the prototype architecture, ordering required electronic components, tracking component delivery status, and preparing the physical prototype model. After receiving the components, the circuit was assembled and tested, resulting in a **successfully completed working prototype with a verified circuit model and functional embedded code**. Regular progress updates were reported to the projectmentor.
- In parallel, research was conducted on **integrating Large Language Models (LLMs) with embedded hardware**, focusing on understanding how intelligent features can be added to the pill box system for future enhancements such as smart assistance, reminders, and data interpretation.
- **Your Role in this Task:** - Embedded hardware developer and researcher responsible for prototype preparation, circuit assembly, functional testing, progress reporting, and exploratory research on LLM integration with embedded systems

- **.What new you do in this Task:** - I contributed to completing the **end-to-end physical prototype** of the American-style QR pill box, starting from component procurement to circuit assembly and code validation. Additionally, I explored the feasibility of adding **AI/LLM-based intelligence** to embedded hardware, extending the project scope beyond basic sensing and communication.
- **What new thing you learn from this Task:** - Learned practical aspects of **hardware prototyping**, including component dependency management, real-time project delays, and iterative testing. Gained understanding of how **LLMs can be integrated indirectly with embedded systems** through gateways, cloud platforms, or companion devices, and how such integration can enhance smart healthcare products.

- **Task Assign No:** - 3
- **Task Assign Month:** - August
- **Team Name:** - Embedded System Development
- **Task Name:** - PCB Design, Hardware Verification, and Research on LLM/SLM Integration for American Version QR Pill Box
- **Task Assign By:** - Avi Kulkarni Sir

- **Work Description:** - Worked on **PCB hardware design, verification, and optimization** for the American version of the Smart QR Pill Box as part of the Pan Health Embedded Development team. The tasks included preparing **PCB layouts for open and closed circuit configurations**, adjusting PCB dimensions to fit enclosure and hardware constraints, and performing iterative design modify accuracy.

The PCB design was finalized and submitted to the college for fabrication, with coordination on expected delivery timelines. Parallel to PCB development, participated in **research activities focused on identifying suitable hardware platforms that can support Small Language Models (SLM) / Large Language Model (LLM) integration** with embedded systems. This research involved studying how intelligent models can be injected into hardware prototypes either directly or through external gateways, cloud support, or companion devices. Information and findings were shared among teammates, and insights were gathered from team members actively working on LLM/SLM hardware compatibility.

- **Your Role in this Task:** - PCB designer and hardware verification engineer, actively involved in layout preparation, dimensional optimization, design submission, and technical coordination. Also contributed as a research participant by gathering, understanding, and sharing information related to LLM/SLM integration with embedded hardware.

- **.What new you do in this Task:** - I refined and finalized PCB designs by making dimension-level adjustments suitable for real hardware fabrication and enclosure fitting. I also supported the team by connecting PCB-level design decisions with future **AI-enabled hardware expansion**, ensuring that the design remains scalable for LLM/SLM integration.
- **What new thing you learn from this Task:** - Learned practical aspects of **PCB hardware lifecycle**, including layout optimization, fabrication readiness, submission workflows, and dependency on external timelines. Gained conceptual understanding of how **LLM/SLM capabilities can be integrated into embedded products**, the hardware limitations involved, and the importance of designing PCBs with future intelligence and scalability in mind.

- **Task Assign No:** - 4
 - **Task Assign Month:** - August
 - **Team Name:** - Embedded System Development
 - **Task Name:** - PCB Hardware Checking, Manual PCB Fabrication, and Research on PCB Design Methods for American Version Pill Box
 - **Task Assign By:** - Avi Kulkarni Sir
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- **Work Description:** - Worked extensively on **PCB hardware checking and preparation for the American version Smart Pill Box**. The tasks involved designing PCB models for the pillbox, verifying hardware layout suitability, and identifying required modifications such as **microcontroller mounting holes**. Along with software-based PCB design, hands-on work was carried out using **copper sheet-based manual PCB fabrication techniques**. This included printing PCB layouts on photo paper, transferring the layout onto a copper board using heat (iron box method), and etching excess copper using **Ferric Chloride (FeCl₃)** solution.
 Additionally, conducted research on **various PCB design and fabrication methods**, explored learning models and tools for PCB design, and evaluated alternate PCB preparation techniques. PCB designs were finalized and submitted to the college for further processing and fabrication approval.
 - **Your Role in this Task:** - PCB hardware checking engineer and fabrication assistant, responsible for PCB model preparation, manual PCB fabrication, layout verification, research on PCB learning models, and submission coordination.

- **.What new you do in this Task:** - I actively participated in **manual PCB fabrication**, applying real-world etching and layout transfer techniques instead of relying only on software simulations. I also verified PCB readiness for microcontroller integration and explored multiple PCB design methods to improve accuracy and learning depth.
- **What new thing you learn from this Task:** - Learned the **complete practical PCB fabrication workflow**, including copper board preparation, toner transfer method, chemical etching using FeCl_3 , and post-etch inspection. Gained understanding of PCB hardware limitations, mounting considerations, and different approaches to learning and implementing PCB design for embedded healthcare products.

- **Task Assign No:** - 5
- **Task Assign Month:** - September
- **Team Name:** - Embedded System Development
- **Task Name:** - PCB Assembly, ESP32 Integration, Hardware Testing, and Real-Time Validation for American Version Smart Pill Box
- **Task Assign By:** - Avi Kulkarni Sir
- **Work Description:** - Worked on the **final stages of PCB hardware development and validation** for the American version Smart Pill Box. The tasks included completing the full PCB model, creating required **mounting holes for the ESP32 DevKit V1**, by **soldering the ESP32 module onto the PCB board**.

To enable pill detection functionality, **Aluminium foil was prepared and placed at the bottom of the pillbox** to act as a conductive layer interfaced with the PCB circuitry. After completing the PCB assembly, the focus shifted to **real-life testing and verification** of the PCB that the circuit works correctly outside simulation environment.

The completed PCB hardware and working model were demonstrated to the project mentor, and further testing activities were carried out, including recording output videos and validating the system performance in practical conditions.

- **Your Role in this Task:** - PCB assembly and hardware testing engineer responsible for PCB completion, ESP32 soldering, conductive interface preparation, mechanical mounting adjustments, real-world testing, and demonstration of the working model.

- **.What new you do in this Task:** - I transitioned the project from **PCB fabrication to full hardware assembly and testing**, integrating the ESP32 directly onto the board and validating the design under real operating conditions. I also implemented a **conductive detection mechanism using aluminium foil**, bridging the mechanical pillbox structure with electronic sensing.
- **What new thing you learn from this Task:** - Learned hands-on skills in **PCB assembly and soldering**, ESP32 hardware mounting considerations, and the importance of mechanical–electrical integration in embedded products. Gained experience in **real-time hardware debugging, functional validation, and demonstration** of embedded systems, especially for healthcare-oriented devices.

- **Task Assign No:** - 6
 - **Task Assign Month:** - September
 - **Team Name:** - Embedded System Development
 - **Task Name:** - PCB Hardware Assembly, Demonstration, Extension Planning, and Real-Time Testing for Smart Pill Box
 - **Task Assign By:** - Avi Kulkarni Sir
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- **Work Description:** - Worked on **assembling the PCB hardware** for the Smart Pill Box project as part of the Pan Health Embedded System Development team. The activities included completing the PCB design, assembling the PCB model, and **soldering the ESP32 module onto the PCB board**. A conductive interface was prepared using **aluminium foil at the bottom of the pillbox** to support pill detection functionality.
 After assembly, the PCB hardware was **tested in real-life operating conditions** to verify correct functionality. The completed PCB design and working model were demonstrated to the project mentor, and the operational flow was clearly explained.
 In addition to hardware assembly, research was carried out on **PCB design extensions**, including exploring suitable extensions and improvements to enhance the PCB layout, scalability, and future feature integration.
 - **Your Role in this Task:** - PCB hardware assembly engineer responsible for assembling PCB components, ESP32 soldering, conductive interface preparation, functional testing, demonstration to the mentor, and research on PCB design extensions.

- **.What new you do in this Task:** - I completed the **full hardware assembly cycle**, from PCB model completion to real-time testing and demonstration. I also initiated exploration of **PCB extensions and design improvements**, linking current hardware functionality with future expansion needs.
- **What new thing you learn from this Task:** - Learned practical skills in **PCB hardware assembly, soldering, and real-world testing**. Gained experience in explaining technical demonstrations to a mentor and understanding the importance of **PCB extensions, scalability, and iterative improvement** in embedded system product development.

- **Task Assign No:** - 7
- **Task Assign Month:** - October
- **Team Name:** - Embedded System Development
- **Task Name:** - 3D Design Development, PCB Hardware Assembly Preparation, and Component Research for American Version Smart Pill Box
- **Task Assign By:** - Avi Kulkarni Sir

- **Work Description:** - Worked on the **mechanical and hardware preparation phase** of the American version Smart Pill Box project. The primary focus was on **designing the 3D model of the pillbox**, especially the **base and bottom structure**, to ensure proper fitting and support for the PCB and electronic components. Multiple iterations of the 3D design were carried out to refine the base structure and improve compatibility with the embedded hardware. In parallel, activities related to **PCB hardware assembly preparation** were performed. This included researching required electronic components, identifying suitable PCBs for the American version of the pillbox, and beginning the assembly process as components became available. Some assembly activities were delayed due to incomplete component availability, and continuous research was carried out to finalize component selection.
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- **Your Role in this Task:** - 3D design contributor and hardware preparation engineer responsible for designing the pillbox base model, supporting PCB integration planning, researching components, and coordinating hardware assembly readiness.

- **.What new you do in this Task:** - I contributed to the **3D mechanical design** of the pillbox by creating and refining the base structure to support PCB mounting and component placement. I also bridged the gap between **mechanical design and electronic assembly** by ensuring alignment between the 3D model and the PCB layout requirements.
- **What new thing you learn from this Task:** - Learned fundamentals of **3D modeling for embedded product enclosures**, including base design, dimensional alignment, and hardware fitment. Gained experience in **component dependency management**, understanding how hardware availability impacts assembly timelines, and the importance of coordinating mechanical design with PCB and electronics integration in product development.

- **Task Assign No:** - 8
- **Task Assign Month:** - October
- **Team Name:** - Embedded System Development
- **Task Name:** - Battery Integration and 3D Design Development for American Version Smart Pill Box
- **Task Assign By:** - Avi Kulkarni Sir

- **Work Description:** - Worked on **integrating a rechargeable battery system** into the Smart Pill Box and continued the **3D design development** for the American version of the pillbox. The task involved selecting and integrating a rechargeable battery along with charging and boost modules such as **TP4060** and **MT3608** to ensure proper power delivery to the system. All required components including battery, battery holder, and modules were collected, and battery setup was successfully completed.

In parallel, significant progress was made on the **3D model design** of the pillbox, focusing on the **bottom structure**, top and bottom layer integration, and additional features like a **phone holder**. The 3D model design was nearly finalized, with printing planned after the college reopened post-Dussehra holidays.

- **Your Role in this Task:** - Responsible for battery integration planning, module selection, component collection, and contributing to the 3D mechanical design of the pillbox enclosure.

- **.What new you do in this Task:** - I integrated a **rechargeable battery system** using TP4060 and MT3608 modules and worked on enhancing the **3D design** by adding top–bottom layers and preparing the enclosure model for real-world printing and assembly.
- **What new thing you learn from this Task:** - Learned practical knowledge about **rechargeable battery integration**, charging and boost converter usage, and power management for embedded systems. Also gained hands-on experience in **3D enclosure design**, dimensional accuracy, and preparing models for physical 3D printing.

- **Task Assign No:** - 9
- **Task Assign Month:** - November
- **Team Name:** - Embedded System Development
- **Task Name:** - Final Assembly and New PCB Design for American Version Smart Pill Box
- **Task Assign By:** - Avi Kulkarni Sir

- **Work Description:** - Worked on **designing a new PCB with 10 pill bay slots** for the American version of the Smart Pill Box and continued the **final assembly process**. The PCB design was completed and the update was shared with the team lead. Parallely, efforts were made to prepare the **final product outcome**, including fixing the PCB, battery, and charging module inside the enclosure using **Anabond gum and glue sticks**.

Mechanical adjustments were carried out by trimming the PCB edges (0.5 mm on all sides) to fit inside the enclosure. Minor dimensional mismatches were identified and addressed. Laser cutting support was planned and coordinated by visiting a laser cutting shop. An **acrylic sheet (blue, shiny finish)** was purchased to enhance the final appearance of the product. The final output video was shown to Avi Sir, and the pillbox achieved a near **final-product look** in terms of color, shape, and finish.

- **Your Role in this Task:** - Designed the new PCB with 10 pill bay slots, participated in final assembly, handled enclosure fitting, coordinated laser cutting work, and assisted in presenting the final output to the mentor.

- **.What new you do in this Task:** - I designed a **new PCB layout with 10 pill bays**, performed real-time **mechanical fitting and trimming**, selected enclosure materials, and actively worked on giving the prototype a **product-level finish**.
- **What new thing you learn from this Task:** - Learned about **mechanical tolerance issues** between PCB and enclosure, practical assembly techniques using adhesives, coordination with **laser cutting processes**, and how product aesthetics (material, color, finish) impact the final prototype quality.

- **Task Assign No:** - 10
- **Task Assign Month:** - December
- **Team Name:** - Embedded System Development
- **Task Name:** - New PCB Design (10-Slot), Product Testing, Explanation, and Courier Coordination
- **Task Assign By:** - Avi Kulkarni Sir
- **Work Description:** - Worked on the **complete design and development of a new 10-slot Smart Pill Box PCB**, starting from schematic design, PCB routing, and preparation of **Gerber files** for manufacturing. The completed PCB design was shared with the team lead for review.

After assembly, the **Smart Pill Box was tested at the office**, including power-up, basic functional checks, and overall working verification. Once testing was completed successfully, the **tested pillbox was demonstrated and submitted to Avi Sir**.

Additionally, explained the **working and operation of the pillbox** for better understanding and user-level clarity. Managed the **courier process** to send the American version of the Smart Pill Box to the Pune office, including dispatch, tracking, and confirmation of successful delivery.

- **Your Role in this Task:** - Responsible for PCB design (10-slot version), product testing at office, explaining pillbox operation, submitting the tested unit to Avi Sir, and coordinating courier delivery to the Pune office.

- **.What new you do in this Task:** - I independently completed the **full PCB design**, prepared Gerber files, performed **office-level testing**, demonstrated the product to Avi Sir, and handled **logistics and delivery coordination**.
- **What new thing you learn from this Task:** - Learned the **complete PCB design-to-testing workflow**, practical **hardware testing and validation**, effective **product demonstration**, and real-world **logistics handling** in embedded product development.

FINAL CONCLUSION INTERNSHIP REPORT

- **Internship Domain:** – Embedded Systems Development
- **Team Name:** – Pan Health Development (Embedded)
- **Project Title:** – Smart QR-Based American Version Pill Box
- **Task Assigned By:** – Avi Kulkarni Sir
- **Intern Role:** – Embedded Hardware Design & Development Intern

- **Work Description : -**

During this internship period, I actively worked on the **end-to-end development of an American version Smart Pill Box**, covering **research, hardware design, PCB development, assembly, testing, and final product delivery**.

The work started with **fundamental research tasks**, including detailed documentation on **Printed Circuit Boards (PCB)**, their types, structure, fabrication methods, and assembly techniques. This foundation helped in understanding real-world PCB design and manufacturing constraints.

I then conducted **battery research for ESP32-based systems**, identifying a **low-cost, rechargeable power solution** suitable for a 3.3V ESP32 controller. After comparing LiPo, Li-Ion, LiFePO₄, and NiMH batteries, a **3.7V LiPo/Li-Ion battery with TP4056 charging module** was finalized. Later, **MT3608 boost module and TP4060 charging controller** were integrated for stable and safe battery operation.

A major phase of the internship involved **working on the American version QR Pill Box prototype**. This included:

- Receiving and organizing hardware components
- Preparing circuit models
- Writing and testing working ESP32 code
- Building and validating hardware prototypes

Parallelly, I worked extensively on **PCB hardware design and checking**, which included:

- Schematic design
- PCB layout and routing
- Dimension adjustments
- Manual PCB fabrication using **copper sheets, photo paper transfer, iron box method, and FeCl₃ etching**
- Drilling holes for ESP32 DevKit V1
- Soldering ESP32 and other components
- Conductivity testing using aluminum foil integration

After completing the PCB, I moved to **PCB assembly and validation**, where real-time hardware testing was done. The PCB was powered, tested for functionality, and verified for real-life usage. The completed and tested PCB was **demonstrated and submitted to Avi Sir** for review.

Further, I worked on **researching hardware suitable for SLM/LLM integration**, understanding constraints related to memory, power, and processing for embedded AI applications.

Another important contribution was **3D design and mechanical assembly** of the pill box:

- Designed top and bottom enclosure models
- Prepared base and slot alignment
- Worked on 3D printing readiness
- Adjusted dimensions for PCB fitting
- Integrated phone holder and pill bay structure

I also contributed to **final product assembly**, including:

- Fixing PCB, battery, and charging modules inside the box
- Using adhesive solutions (Anabond, glue sticks)
- Acrylic sheet selection and laser cutting coordination

- Resolving PCB fitting tolerance issues

In the final phase, I designed a **new PCB for a 10-pill-bay version**, completing:

- Full schematic design
- PCB routing
- Gerber file generation
- Design verification and team lead communication

The **final American version Smart Pill Box** was:

- Fully assembled
- Tested at office
- Demonstrated to Avi Sir
- Packed and couriered to the Pune office

I coordinated the **entire courier process**, including dispatch through DTDC, tracking, and confirmation of successful delivery to the Pune office, where the product was received by Rakesh Sir.

- **Your Role in This Internship:-**

- Primary contributor to PCB design, fabrication, and testing
- Battery selection and power management integration
- Prototype preparation and hardware debugging
- 3D design support and mechanical assembly
- Office-level testing and demonstration
- Final product submission and courier coordination

- **What New You Did in This Internship:**

- Designed and fabricated **multiple PCB versions**, including a **10-slot pill box PCB**
- Performed **manual PCB manufacturing**
- Integrated rechargeable battery systems
- Conducted real-time hardware testing
- Managed product delivery logistics
- Worked on **complete product lifecycle**, from idea to delivery

- **What New Things You Learned:-**

- Complete **PCB design workflow** (schematic → layout → fabrication → testing)
- Battery management and charging circuits
- ESP32 hardware integration
- Mechanical–electrical alignment in product design
- Importance of **testing, documentation, and submission**
- Real-world challenges in embedded product development

- **Final Outcome:**

- Working hardware
- Tested PCB
- Rechargeable power system
- 3D enclosure
- 10-slot PCB design
- Office-tested and submitted product

This internship provided **strong hands-on experience** in embedded systems, PCB design, and real-world product development.

ATTENDANCE PROOF'S

ALL YOUR TASK REPORT ATTACHMENT'S

Complete Set of Internship Task Reports (Overall Folder) Link:

https://drive.google.com/drive/folders/1bs1gUKrbWcgGYnLo6r5ucYx5c3WsFKbo?usp=drive_link

Task Attachments by Month:

Sr.No.	Month	Relevant Tasks (from Task Report)	Specific Google Drive Folder Link
1)	June 2025	Design, Simulation, Sensor Research, PCB Development and Prototype Planning of Smart QR-based Pillbox System	https://drive.google.com/drive/folders/1NVOdGhDcM2oAgK6z4zq-kOZtSRnIBL-?usp=drive_link
2)	July 2025	Development and Prototyping of American Version Smart QR Pill Box with Advanced Features	https://drive.google.com/drive/folders/1zt5bsaez-wSptyRMXOXcFvHHWLkK58T6?usp=drive_link
3)	August 2025	PCB Design, Hardware Verification, and Research on LLM/SLM Integration for American Version QR Pill Box	https://drive.google.com/drive/folders/1ng7G-vA4PRINag4W5zsnxoetsw4HLp5Q?usp=drive_link
4)	August 2025	PCB Hardware Checking, Manual PCB Fabrication, and Research on PCB Design Methods for American Version Pill Box	https://drive.google.com/drive/folders/1ng7G-vA4PRINag4W5zsnxoetsw4HLp5Q?usp=drive_link

Sr.No.	Month	Relevant Tasks (from Task Report)	Specific Google Drive Folder Link
5)	September 2025	PCB Assembly, ESP32 Integration, Hardware Testing, and Real-Time Validation for American Version Smart Pill Box	https://drive.google.com/drive/folders/16_INTgeHFnaN4xd3JEnVH65JLUEczb6H?usp=drive_link
6)	September 2025	PCB Hardware Assembly, Demonstration, Extension Planning, and Real-Time Testing for Smart Pill Box	https://drive.google.com/drive/folders/16_INTgeHFnaN4xd3JEnVH65JLUEczb6H?usp=drive_link
7)	October 2025	3D Design Development, PCB Hardware Assembly Preparation, and Component Research for American Version Smart Pill Box	https://drive.google.com/drive/folders/1FSeSKfrGhRUjetbYlX0J4yNBX_OzXxNI?usp=drive_linkk
8)	October 2025	Battery Integration and 3D Design Development for American Version Smart Pill Box	https://drive.google.com/drive/folders/1MLV94BUD1rdu06kXvH8-Urc2Ft7Fr5Yn?usp=drive_link
9)	November 2025	Final Assembly and New PCB Design for American Version Smart Pill Box	https://drive.google.com/drive/folders/1RYjInrafZnr7xmgSOLVsklwYD23854XQ?usp=drive_link
10)	December 2025	New PCB Design (10-Slot), Product Testing, Explanation, and Courier Coordination	https://drive.google.com/drive/folders/1Kv8J6FJBgcx0jGbIG1gWpG_fbkzzKKU4?usp=drive_link



Internship Offer

StartupWorld™



Congratulations! We are delighted to inform you that you have been selected for an internship position within our organization.

- **Date of Joining (DOJ):** 05 Jun 2025
- **Team Name:** SMCESD
- **Team Leader's Email:** rakesh@blueplanetinfosolutions.com
- **Internship Mode:** Part-Time (Work from Home)
- **Duration:** 6 Month
- **Internship Type:** Unpaid
- **Timing:** 18:00 - 22:00
- **Hours:** 4 Hours/Day
- **Designation:** Embedded Systems Development
- **Responsibility:** Assists in designing, coding, testing, and debugging firmware/software for microcontrollers and embedded devices in real-time environments.

As part of your internship, you will be working remotely from home. The duration of the internship is 6 Month, and you will be designated as a Embedded Systems Development intern.

To receive an internship certificate, we require you to complete the full 6 Month internship period. Kindly submit your reports to your reporting manager for evaluation and confirmation. If you have any questions or need assistance during your internship, feel free to contact us at **8380037181**

We understand the importance of flexibility, and your working hours are adaptable to suit your schedule.

We are excited to have you on board and look forward to your valuable contributions during your internship.

Best regards,
Blue planet info solutions

Rockey

RECIPIENT
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NO OBJECTION LETTER FOR INTERNSHIP
COMPLETION CERTIFICATE ISSUE

INTERNSHIP FEEDBACK FORM

- 1. How would you describe your overall experience working as an intern in Company?**

Answer: My overall experience was fantastic; it was challenging but really satisfying to work on a complete smart device from start to finish.

- 2. How effectively were you able to apply your academic knowledge or skills during your internship?**

Answer: I was very effective; I successfully applied my skills in problem-solving, design, and basic programming for a real-world product.

- 3. Did you receive sufficient guidance and support from your team or supervisor during the internship?**

Answer: Yes, the team and supervisor provided excellent support and clear guidance whenever I needed help.

- 4. What were the most valuable skills, lessons, or experiences you gained during your internship?**

Answer: The most valuable experience was learning how to lead a design project and connect all the physical and electronic parts together smoothly.

- 5. How would you rate the work environment, communication, and culture within your team?**

Answer: The culture was friendly, encouraging, and very focused on collaboration, which made communication easy.

- 6. What challenges did you face during your internship?**

Answer: The biggest challenge was making sure the custom-designed plastic case fit all the small electronic components perfectly inside.

7. Form the journey of your internship how support and encourage you more?

Answer: Being trusted to lead the project and deliver a user-friendly product with new features made me feel very motivated and supported.

8. What are your suggestions or recommendations to improve the internship program in Company?

Answer: I recommend offering more clear instruction sessions on advanced topics like how to keep the power running longer and how to prepare final files for manufacturing.